

EXPERIMENT-3

NAME: FAIYAS N

ROLL NO: 240701135

Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI(voice user interface)

1.Command Line Interface

A Command Line Interface (CLI) allows the user to interact with the system by typing commands and inputs through the keyboard.

It is simple and fast because the user directly enters data without using buttons or graphics.

CLI programs require less memory and system resources compared to graphical applications.

In this experiment it creates the file and open the file. If the file is created then shows this file already exist.

IMPLEMENTATION

```
import sys import os
```

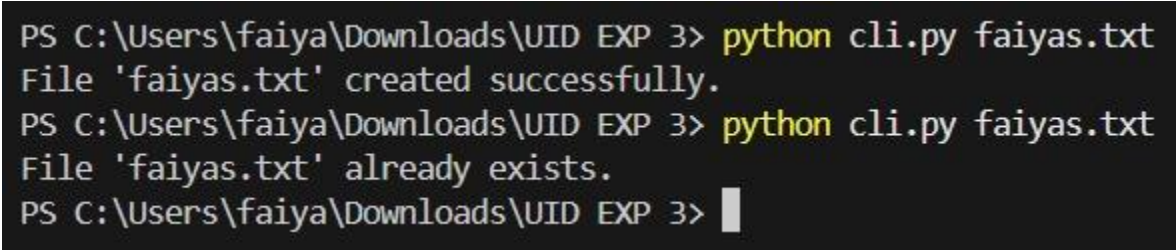
```
def create_file(filename): try:
```

```
    if os.path.exists(filename):
        print(f"File '{filename}' already exists.")
    else:
        with open(filename, 'w') as file:
```

```
        file.write("File created successfully!\n")
        print(f"File '{filename}' created successfully.")
except Exception as e:
    print(f"An error occurred: {e}")
```

```
if name == "main": if len(sys.argv) != 2: print("Usage: python create_file_cli.py
") else: create_file(sys.argv[1])
```

OUTPUT:



```
PS C:\Users\faiya\Downloads\UID EXP 3> python cli.py faiyas.txt
File 'faiyas.txt' created successfully.
PS C:\Users\faiya\Downloads\UID EXP 3> python cli.py faiyas.txt
File 'faiyas.txt' already exists.
PS C:\Users\faiya\Downloads\UID EXP 3> █
```

2.Graphical User Interface

A Graphical User Interface (GUI) allows users to interact with the system using visual elements like windows, buttons, and text boxes instead of typing commands. It is user-friendly and easy to understand because users can click buttons and enter data in input fields.

GUI provides a better visual experience and is suitable for beginners who may not know command-line instructions. In this experiment it creates the file and open the file. If the file is created then shows this file already exist.

IMPLEMENTATION:

```
import tkinter as tk
```

```
from tkinter import messagebox
```

```
import os
```

```
def create_file():
```

```
    filename = filename_entry.get().strip()
```

```
    if not filename:
```

```
        messagebox.showwarning("Warning", "Please enter a filename!")
```

```
        return
```

```
    if os.path.exists(filename):
```

```
        messagebox.showinfo("Info", f"File '{filename}' already exists.")
```

```
        return
```

```
    try:
```

```
        with open(filename, 'w') as file:
```

```
            file.write("File created successfully!\n")
```

```
    messagebox.showinfo("Success", f"File '{filename}' created  
successfully.")
```

```
except Exception as e:
```

```
    messagebox.showerror("Error", f"An error occurred: {e}")
```

```
root = tk.Tk()
```

```
root.title("File Creator")
```

```
root.geometry("400x150")
```

```
root.resizable(False, False)
```

```
tk.Label(root, text="Enter Filename:", font=("Arial", 12)).pack(pady=10)
```

```
filename_entry = tk.Entry(root, width=40)
```

```
filename_entry.pack(pady=5)
```

```
tk.Button(root, text="Create File",
```

```
    bg="blue",
```

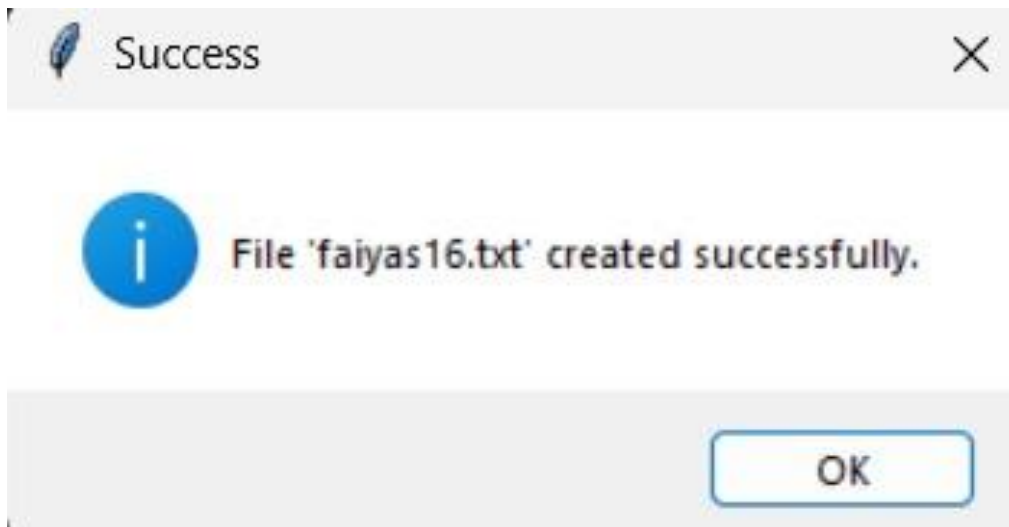
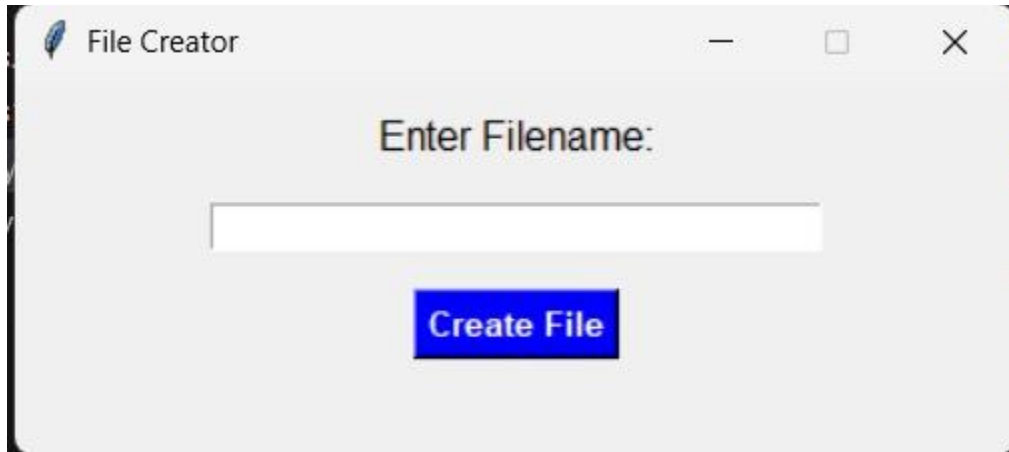
```
    fg="white",
```

```
    font=("Arial", 10, "bold"),
```

```
    command=create_file).pack(pady=10)
```

```
root.mainloop()
```

OUTPUT:



3.Voice User Interface:

```
import speech_recognition as sr import os

def create_file_from_voice(command): words = command.lower().split()

if "create" in words and "file" in words:
    try:
        filename = words[words.index("file") + 1]
```

```

        if os.path.exists(filename):
            print(f"File '{filename}' already exists.")
        else:
            with open(filename, "w") as f:
                f.write("File created using voice
command.\n")

            print(f"✅ File '{filename}' created
successfully.")

    except IndexError:
        print("Please say: Create file filename.txt")

else:
    print("Invalid command. Say: Create file filename.txt")

def listen():
    recognizer = sr.Recognizer()
    mic = sr.Microphone()

    print("🎧 Listening... Say: Create file filename.txt")

    with mic as source:
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
        print("You said:", command)
        create_file_from_voice(command)

    except sr.UnknownValueError:
        print("Sorry, could not understand audio.")
    except sr.RequestError:
        print("Speech Recognition service error.")

```

```
if name == "main": listen()
```

OUTPUT:

```
PS C:\Users\faiya\Downloads\UID EXP 3> python vui.py
Traceback (most recent call last):
  File "C:\Users\faiya\Downloads\UID EXP 3\vui.py", line 1, in <module>
    import speech_recognition as sr
ModuleNotFoundError: No module named 'speech_recognition'
PS C:\Users\faiya\Downloads\UID EXP 3> |
```